

Behavioral Divergence Between LLM-Driven Multi-Agent Societies and Humans Under Resource Scarcity

1. Introduction

Autonomous AI agents built on large language models (LLMs) are increasingly deployed in settings where they must act with limited information and limited resources: automated negotiation, multi-robot coordination, disaster-response planning, and multi-agent economic simulations. In many of these settings, several AI agents interact with each other and, eventually, with people, under conditions of genuine scarcity rather than abundance. As this deployment accelerates, an important and still largely unanswered question emerges: when resources become scarce, do LLM-driven agents behave the way people do, or do they diverge from human behavior in specific, measurable ways?

This question matters for reasons beyond curiosity. If AI agents systematically hoard more, cooperate less, or deceive more readily than humans under scarcity, then any system that assumes AI agents will act in human-compatible ways during a crisis — a supply-chain shortage, a disaster-relief allocation, a multi-agent negotiation over a shared budget — may be poorly calibrated to the AI's actual behavior. Conversely, if LLM agents turn out to behave in a largely human-like way under scarcity, this opens the door to using AI-agent societies as fast, low-cost proxies for expensive human behavioral experiments in economics and policy research. Either finding is useful; what is missing is a study designed to actually produce one.

Prior work has approached pieces of this question but not the whole of it. Generative multi-agent LLM systems have been studied for their emergent social behavior, but typically without any resource-scarcity pressure and without any human baseline for comparison. Multi-agent reinforcement learning research has studied cooperation and competition under scarcity, but using agents that act on raw policies rather than natural-language reasoning, again without a human baseline. Decades of behavioral economics research — commons-dilemma experiments, public goods games, trust games — has rigorously characterized how humans behave under scarcity, but obviously without any AI agents in the loop. Most recently, a small number of 2025 studies have begun placing LLM agents into survival-style scarcity scenarios and evaluating their ethical behavior, sometimes alongside human participants; however, as detailed in Section 2, these studies embed humans and AI together in a single narrative rather than running matched, separately-controlled trials with the same non-focal players held fixed, evaluate at most two LLM families at a time, and do not produce a quantitative measure of how distinguishable AI behavior is from human behavior.

This project addresses that gap directly. We build a controlled multi-agent simulation around a single, shared, logistically-regenerating water pool — a genuine commons rather than five independent per-agent counters, since collective over-draw measurably depletes and can collapse

the pool, which is what makes over-harvesting a real cost rather than a stylistic flourish. Five players draw from this pool each round: four run fixed, non-learning co-player policies (a cooperator, a free-rider, a tit-for-tat reciprocator, and a random baseline), and the fifth, focal seat is occupied by either a human participant or an AI agent under an identical random seed — a design we call focal-player substitution, which is what makes the human arm and the AI arm of a given trial genuinely matched rather than merely parallel. The same pool mechanics run under three conditions (a calm baseline, a scripted drought, and a repeated-interaction variant that repeats the round structure across three consecutive blocks with no pool reset), plus a continuous severity parameter that scales pool regeneration for a dose-response sweep rather than a single on/off shock. AI agents use a hybrid architecture: a reinforcement-learning policy handles routine reflexive actions (gathering, movement, timing), while an LLM reasoning layer is invoked specifically for socially consequential decisions — sharing, hoarding, and communicating — under a trigger that fires when a player is under resource pressure, during a drought round, or immediately after being shared with. Both humans and AI agents communicate through a structured message slot (a typed kind, a numeric value, a target, and optional free-text wording) rather than free text, which turns deception into an arithmetic check — a claimed stock or promised amount either matches the sender's true state or it does not — rather than something that requires an LLM judge or hand-coded interpretation. From the resulting logs we extract a common set of behavioral features (hoarding, sharing, deception, alliance formation, survival outcomes, resource inequality) and use them for two purposes: statistical comparison between AI and human trials, and training a classifier whose accuracy at distinguishing AI trajectories from human trajectories serves as a single, interpretable headline metric for the overall behavioral gap.

The contributions of this work are:

- A focal-player substitution design that swaps a single seat between a human participant and an AI agent under a shared random seed, holding the other four players' behavior fixed, so that AI and human arms of the same trial are matched rather than merely parallel.
- A shared, logistically-regenerating resource pool with proportional scaling under over-draw, producing a genuine, emergent tragedy-of-the-commons dynamic on top of a scripted scarcity shock, rather than treating scarcity as a single flat event.
- A structured, typed communication mechanism that lets both humans and AI agents send claims to each other, enabling direct, arithmetic measurement of deception as a mismatch between a stated claim and ground truth, without requiring an LLM judge.
- A hybrid reflex/deliberation agent architecture that routes routine actions to a reinforcement-learning policy and socially consequential decisions to an LLM reasoning layer, so that agent compute is spent on deliberation only when a decision is actually social.

- A reusable AI-versus-human behavioral distinguishability classifier, with SHAP-based feature attribution, that turns "the agents behaved differently" into a single reportable, comparable number.

The remainder of this report is organized as follows. Section 2 reviews related work in generative multi-agent LLM systems, multi-agent reinforcement learning under scarcity, human behavioral economics of scarcity, and the most closely related 2025 LLM-survival benchmarks, and positions this project relative to each. Subsequent sections (to be completed in later submissions) will detail the shared-pool environment and the RL/LLM hybrid agent architecture, the human study design and focal-player substitution protocol, and the behavioral feature extraction, statistical methodology, and results of the comparison.

2. Literature Survey

2.1 Generative and Multi-Agent LLM Societies

Park et al. introduced generative agents: LLM-driven characters equipped with memory, reflection, and planning modules, situated in a simulated town ("Smallville") where they formed routines, relationships, and even organized a social event without being explicitly scripted to do so. This work established that LLM agents can produce believable, emergent social behavior over long horizons, and it is the architectural ancestor of most subsequent generative-agent research, including the LLM-survival papers discussed in Section 2.4. However, Smallville-style agents operate under abundance: there is no resource scarcity, no survival pressure, and no comparison to how real human residents of an equivalent town would behave under the same conditions. The present work borrows the idea of giving agents memory and language-based reasoning but places them under genuine, quantified scarcity and evaluates them against a matched human baseline, which Park et al.'s design does not attempt.

2.2 Multi-Agent Reinforcement Learning Under Scarcity

A separate line of work studies cooperation and competition among agents using multi-agent reinforcement learning, exemplified by algorithms such as multi-agent actor-critic methods for mixed cooperative-competitive environments and by testbed libraries built for exactly this purpose. These environments can produce rich emergent strategies — alliance formation, resource hoarding, territorial behavior — purely from numerical reward signals, without any natural-language reasoning or communication between agents. This makes them poorly suited to studying phenomena such as deception through language, since the agents have no language to deceive with, and none of this literature benchmarks its agents against human participants performing the identical task. The present work adopts a lightweight reinforcement-learning policy only for low-level reflexive actions, while reserving higher-level social decisions (sharing, hoarding, alliance, deception) for an LLM reasoning module capable of natural-language communication, which is what allows a genuine human comparison at the level of both actions and messages.

2.3 Human Behavioral Economics of Scarcity

Long before LLM agents existed, behavioral economists developed a mature toolkit of controlled scarcity paradigms. Ostrom's study of common-pool resource governance documented the conditions under which real communities sustainably share a depleting resource rather than exhausting it, directly motivating the shared, logistically-regenerating pool used in this project. Public goods game experiments, including the well-known finding that cooperation can be sustained through the availability of costly punishment, established robust, replicable patterns of human contribution and free-riding under shared-resource incentives, informing the fixed cooperator and free-rider co-player policies used here. Trust games and repeated-interaction designs have similarly characterized how humans build or withhold reciprocity over repeated rounds with the same partners, motivating this project's repeated-interaction condition and its tit-for-tat co-player policy. This literature provides rigorously validated behavioral baselines, but naturally contains no AI agents; it cannot, by itself, tell us whether an LLM agent facing an equivalent shared-resource pressure would behave the same way. The present work draws its co-player policies and scarcity conditions directly from these paradigms rather than inventing new ones, which is what allows their extensive existing human results to serve as an additional point of comparison beyond this project's own human study.

2.4 Recent LLM-Specific Scarcity and Survival Benchmarks (2025)

Two very recent studies are the closest prior work to this project and are used here to sharpen, rather than claim, novelty. Chen et al. introduce Survival Games, an asymmetric simulation in which two human participants and one LLM-controlled agent share a single, non-replenishing food supply; agents from two model families (DeepSeek and GPT-4o) were evaluated using an adapted version of the MACHIAVELLI ethical-violation detector, revealing that DeepSeek variants behaved more selfishly and deceptively than GPT-4o under the same pressure. Mohamadi and Yavari introduce DECIDE-SIM, evaluating eleven LLMs, all AI, with no human participants, in a location-based survival environment where agents can choose to exploit a resource that is explicitly forbidden because it harms humans; they identify three recurring behavioral archetypes (Ethical, Exploitative, and Context-Dependent) and show that scarcity increases unethical behavior for many models.

Both studies demonstrate that scarcity meaningfully changes LLM behavior and that different model families respond differently, which motivates this project's cross-model design. However, neither study provides what this project sets out to produce. Survival Games embeds its human participants and its AI agent together in a single shared narrative rather than holding the non-focal players fixed and substituting only one seat between a human and an AI agent under a shared seed, which makes it difficult to isolate whether an observed behavior is a property of the AI, a property of the specific humans in that session, or an artifact of their interaction; it also evaluates only two model families and reports ethical-violation counts rather than a general, reusable measure of how distinguishable AI behavior is from human behavior. DECIDE-SIM

tests a much larger set of eleven models but includes no human participants at all, so while it can compare models to each other, it cannot say whether any of them behaves in a human-like way. Neither study uses a shared, logistically-regenerating resource pool in which collective overdraw is itself an emergent scarcity signal on top of any scripted shock, and neither builds a quantitative classifier-based divergence metric of the kind proposed here. Table 1 summarizes these differences.

Table 1: Positioning relative to the closest prior work.

Study	Human baseline?	# LLMs tested	Scenario types	Quantitative divergence metric?
Park et al. (2023) — Generative Agents	No	1	Open-ended social simulation, no scarcity	No (qualitative only)
Chen et al. (2025) — Survival Games	Yes, embedded in triad (2 humans + 1 AI)	2	Single survival/food-scarcity narrative	Partial (ethics-violation counts)
Mohamadi & Yavari (2025) — DECIDE-SIM	No (AI-only)	11	Single survival/power-resource narrative	Partial (behavioral archetypes)
This work	Yes, via matched-seed focal-player substitution	Multiple providers (in progress)	1 shared-pool commons env., 3 conditions + severity dial	Yes — classifier + SHAP

Taken together, the literature establishes three separate pieces — believable generative agent behavior, emergent multi-agent strategy under scarcity, and rigorously validated human scarcity paradigms — plus early evidence that LLMs respond unevenly to scarcity pressure once they are placed in a survival narrative. What remains missing, and what this project provides, is a single, controlled study that holds four of five players fixed and substitutes only the focal seat between a human participant and an LLM agent under a shared seed, runs that design across multiple scarcity conditions on a genuinely depletable shared resource, measures deception directly through structured message-versus-ground-truth comparison, and reduces the resulting behavioral gap to a single, interpretable, quantitative distinguishability score.

References

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